

Prop transformada fourier derivada n

Proposición 1. Sea $k \in \mathbb{N}$ y $f \in \mathcal{C}^k(\mathbb{R})$ tal que $\forall j \in \mathbb{N}_k \cup \{0\} : f^{(j)} \in \mathcal{C}_0(\mathbb{R}) \cap \mathcal{L}^1(\mathbb{R})$

$$\implies \forall j \in \mathbb{N}_k : \left(f^{(j)}\right)^\wedge(\xi) = (2\pi i \xi)^j \cdot f^\wedge(\xi).$$

Demostración: Se prueba por inducción en j usando el `lem-transformada-fourier-derivada`/Lema 1
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